

COMPUTER SYSTEMS ARCHITECTURE

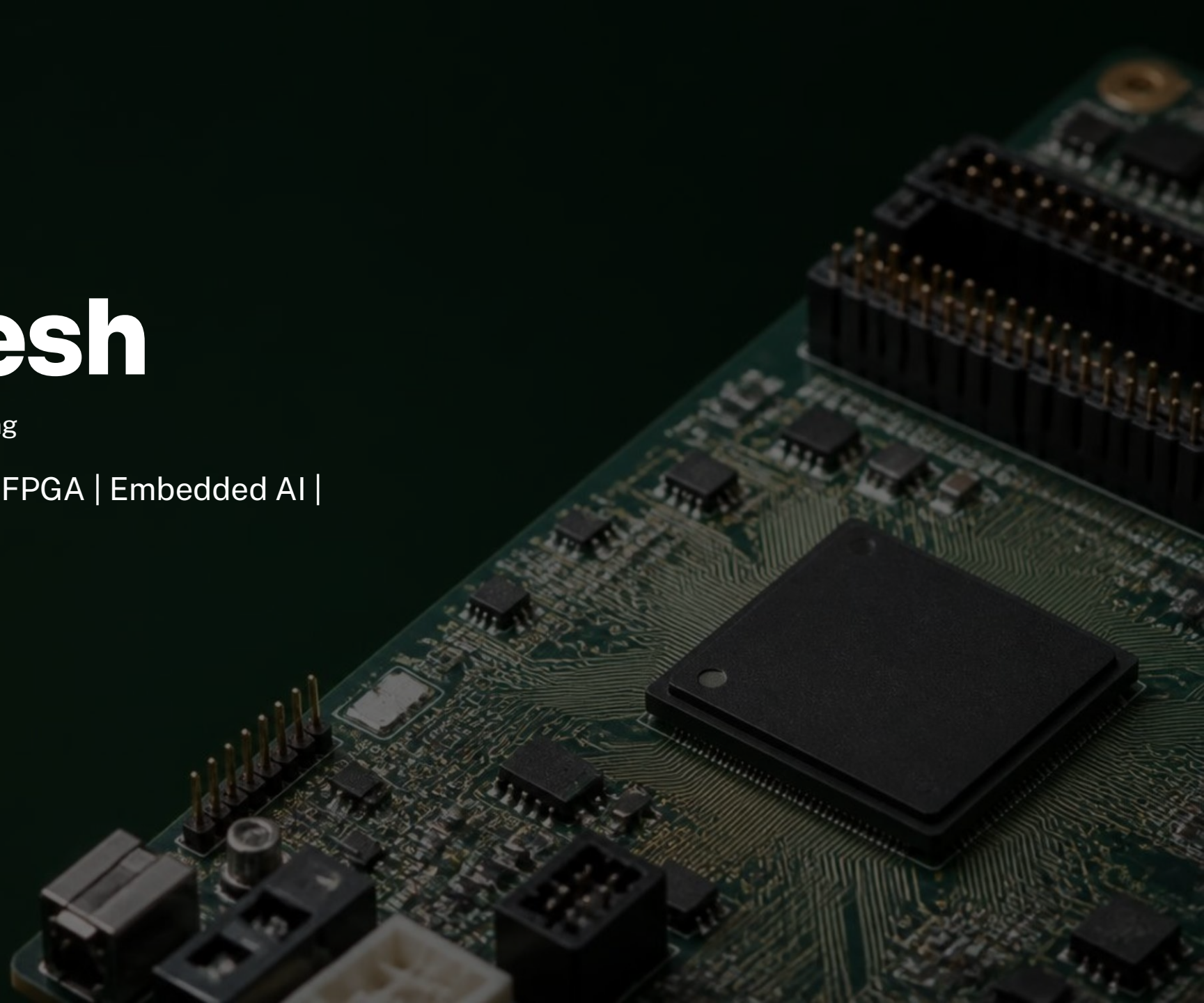
Yahya Iranmanesh

Ph.D. Candidate in Computer Engineering

Hardware Processor for IQWG | FPGA | Embedded AI |
Automation



Instructor, Computer Engineering
Azad University, North Tehran Branch



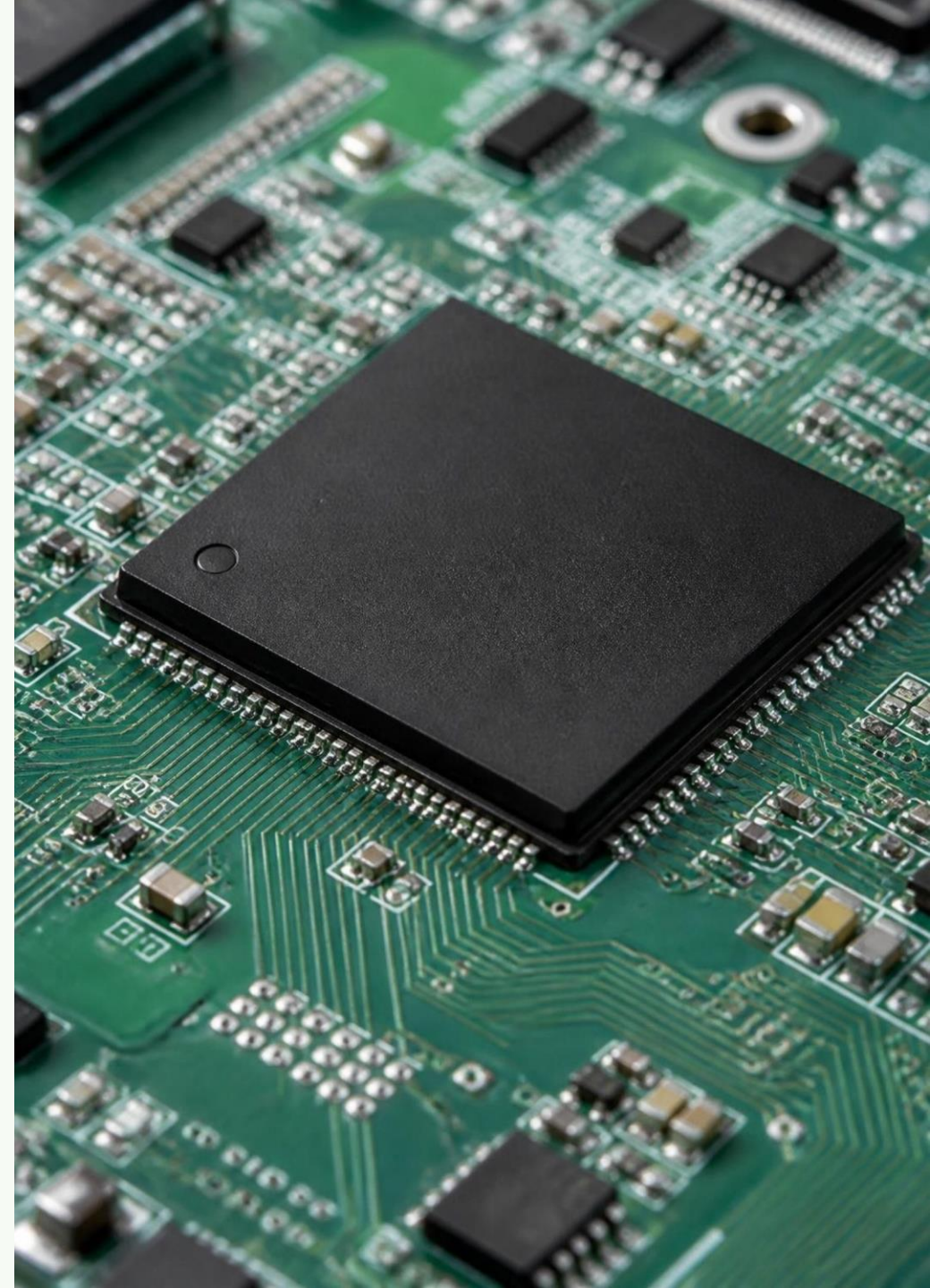
ACADEMIC PROFILE

IQWF Processing need hardware built for them

Instructor, Computer Engineering, Azad University, North
Tehran Branch

Ph.D. candidate in Computer Systems Architecture

Doctoral focus: hardware accelerators for neural networks



RESEARCH AREAS

Five research areas, one goal: efficient neural network hardware

Hardware acceleration

Specialized architectures improve throughput, latency, and power consumption.

FPGA-based prototyping

Programmable hardware enables rapid validation of new architectural concepts.

IQWG design

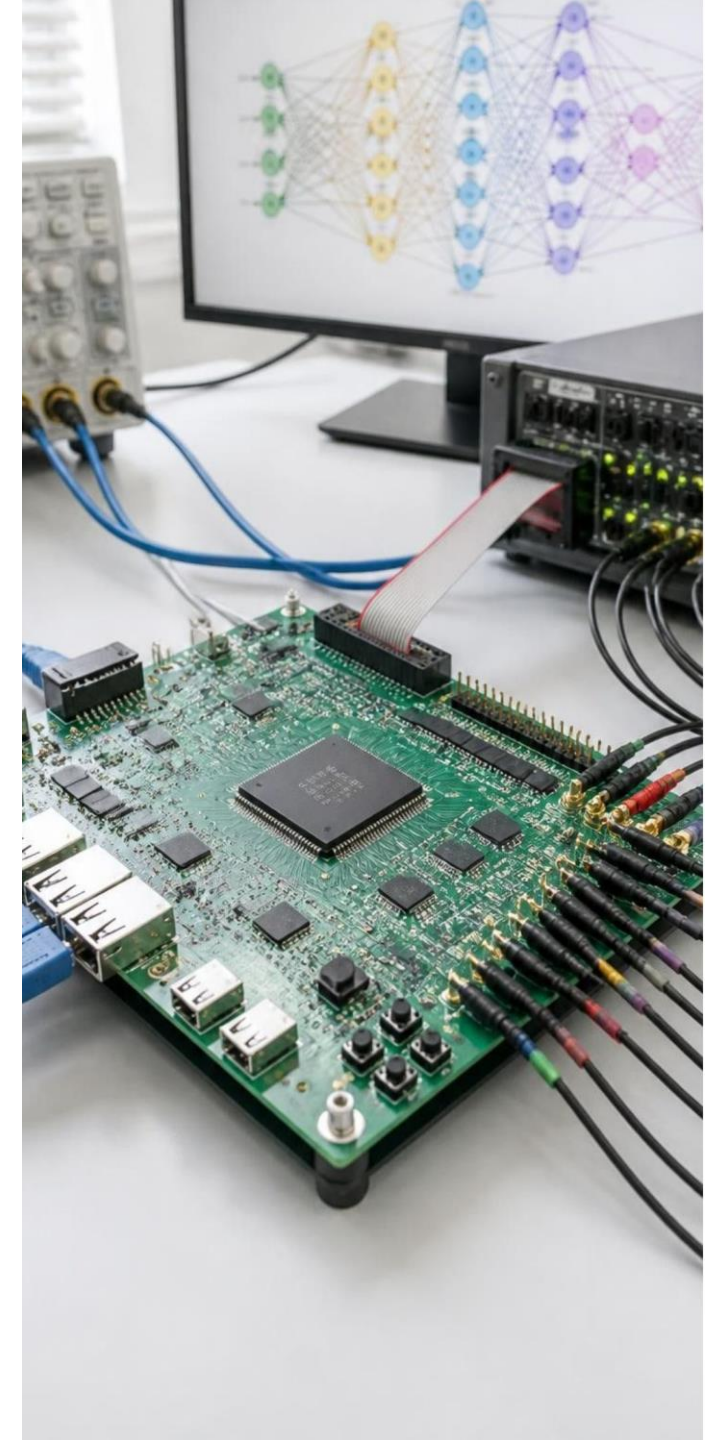
Custom IQWG Processing Units use parallelization and quantization, which reduces numerical precision.

Hardware-software co-design

Hardware constraints and software requirements are balanced together.

Embedded AI

Real-time models serve constrained devices in robotics, machine vision, and automation.



Making AI run on hardware with limited resources

From algorithms to real-time platforms

- Doctoral research develops advanced hardware Processing for IQWG architectures.
- Architectures are tuned for performance, power efficiency, and scalability.
- The work bridges high-performance algorithms and resource-constrained platforms for real-time use.

ACADEMIC ROLE

Computer Systems Architecture

Ph.D. candidate in Computer Engineering, specializing in Computer Systems Architecture.

Instructor at Azad University since February 2023.



RESEARCH PRIORITIES

Research: AI on resource-limited hardware

Hardware-aware neural network optimization, grounded in the constraints of embedded devices.

Performance

Efficient execution of neural network inference.

Power efficiency

Useful computation within a limited energy budget.

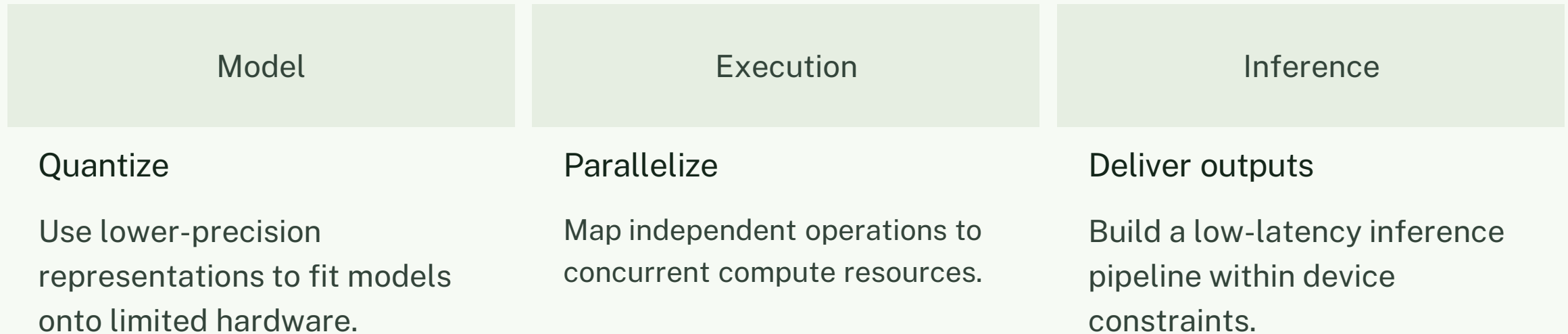
Scalability

Architectures that accommodate changing workloads.

Hardware-software co-design for efficient, reliable embedded systems.

Parallelization and quantization cut inference latency

Two complementary levers: reduce the work each operation carries and execute independent work concurrently.



Conceptual pipeline. Precision, concurrency and hardware resources must be considered together.

Current projects target low-latency, hardware-aware inference

Optimize the architecture

Novel architectures support hardware-aware neural network optimization.

Reduce inference latency

Low-latency inference pipelines target real-time AI applications.

Adapt embedded networks

Parallelization and quantization strategies tailor neural networks to embedded hardware.

Embed AI in control

AI-embedded systems support industrial automation and autonomous control.



APPLIED ENGINEERING

Industrial automation in practice

Production lines

Fully automated production lines, integrating machines and control.

Machine conversion

Converting manual machines into automated systems.

Intelligent control

Intelligent robotics, machine vision and autonomous control.



APPLIED ENGINEERING

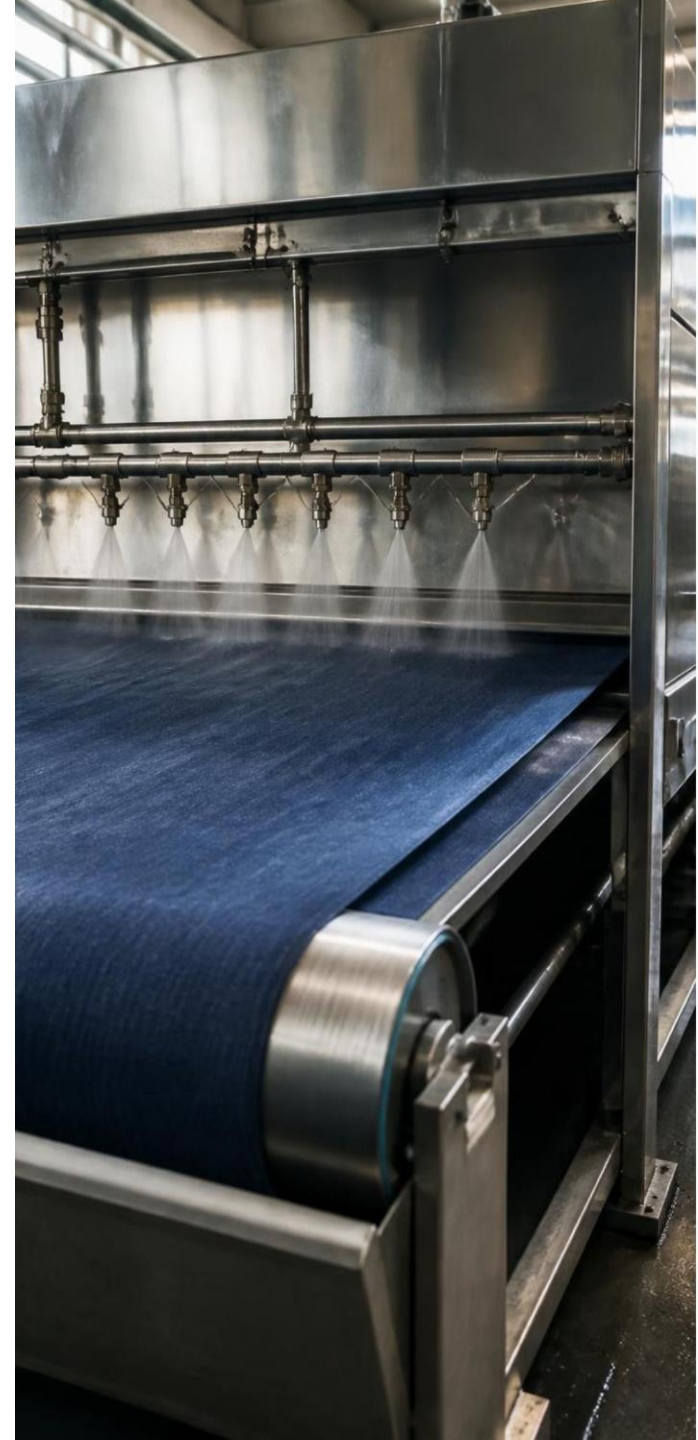
Research reaches the factory floor

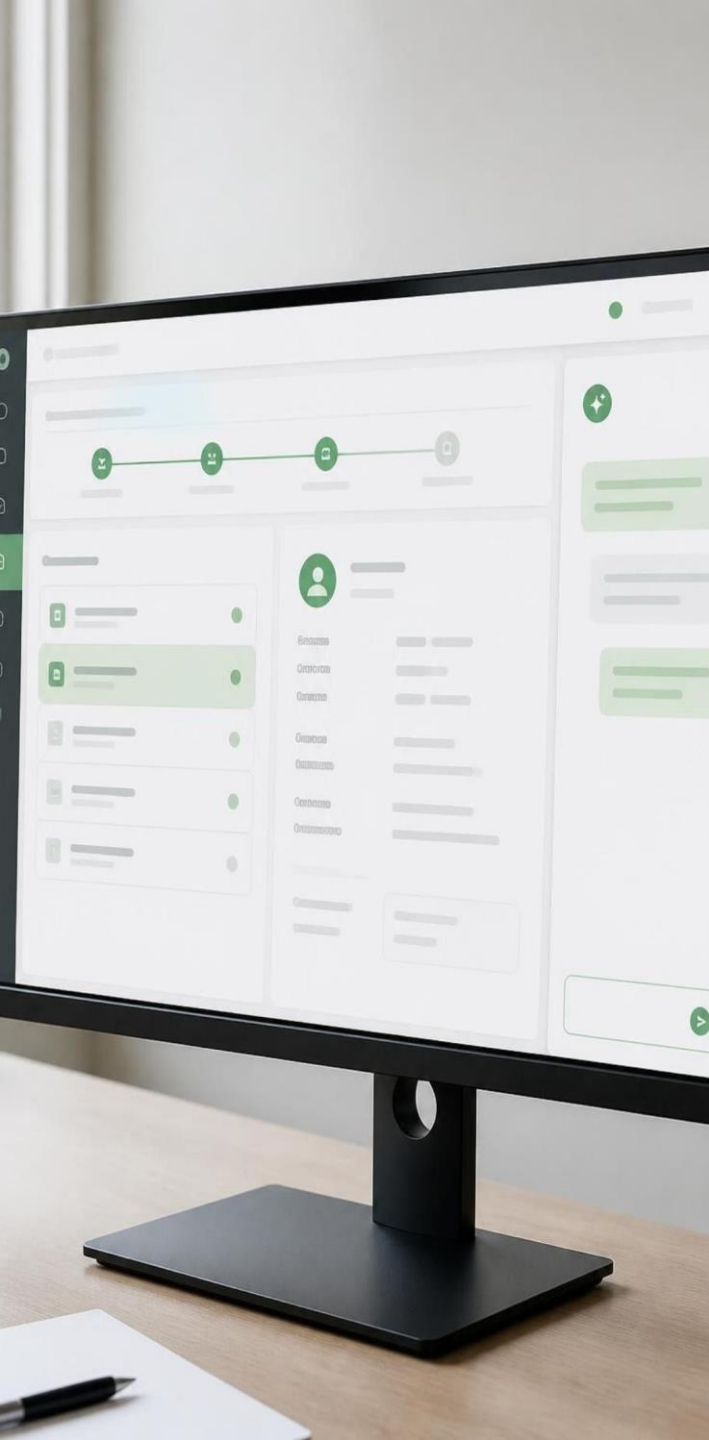
- Designs fully automated production lines and converts manual machines to automated systems.
- Applied work spans intelligent robotics, machine vision, industrial automation, and autonomous control systems.
- Model design and hardware-software co-design work together for efficiency and reliability in industrial settings.

PHA CO.

Nine automation systems delivered with PHA Co.

- IT Manager and Team Leader, July 2007 to present.
- Embossing and liquid spraying on leather (Kurdistan)
- synthetic leather line (Isfahan)
- Photo paper production line (Tehran)
- chemical mixing, filling, and packaging (Pardis)
- Spray fabric dyeing system (Damavand)
- Camper intelligent brake system, CNC robotic arm, smart classroom (Tehran)
- Adaptive cruise control based on depth-of-field image processing (Tehran)





SOFTWARE AND INFRASTRUCTURE

AI-driven software runs whole offices

Full office automation platforms

CRM, workflow management (BPMS), accounting, treasury, and attendance.

Intelligent business processes

Intelligent process automation, digital assistants, and sales optimization.

Network integration at Vanak Co.

Firewalls, routers, VOIP PBX integration with ISDN, and E1 interfaces.

Earlier enterprise systems experience

5 years at Kaveh Glass Industry Group on inventory, sales, purchasing, and production systems.

ENGINEERING FOUNDATIONS

Technical toolkit

HARDWARE

Zynq FPGA · VHDL/Verilog · SoC/NPU · VLSI · ESP32 · AVR · PIC · ARM · PLC/HMI

SOFTWARE

C/C++ · C# · Python · SQL Server · Flutter · REST APIs

SYSTEMS

Linux · Windows · networks · VOIP/Asterisk

REGISTERED INVENTIONS

Two registered patents



Hearing Aid and Modular Battery-Free Wireless Earbuds

Patent #112871, August 2025



Powerful Mechanical Wheelchairs Portal Converter

Patent #96383, February 2018

Both registered with the Iranian Research Organization for Science, Technology and Office of Industrial Property.

Images illustrate device categories, not the patented designs.

Teaching theory through hands-on hardware

Hardware and architecture

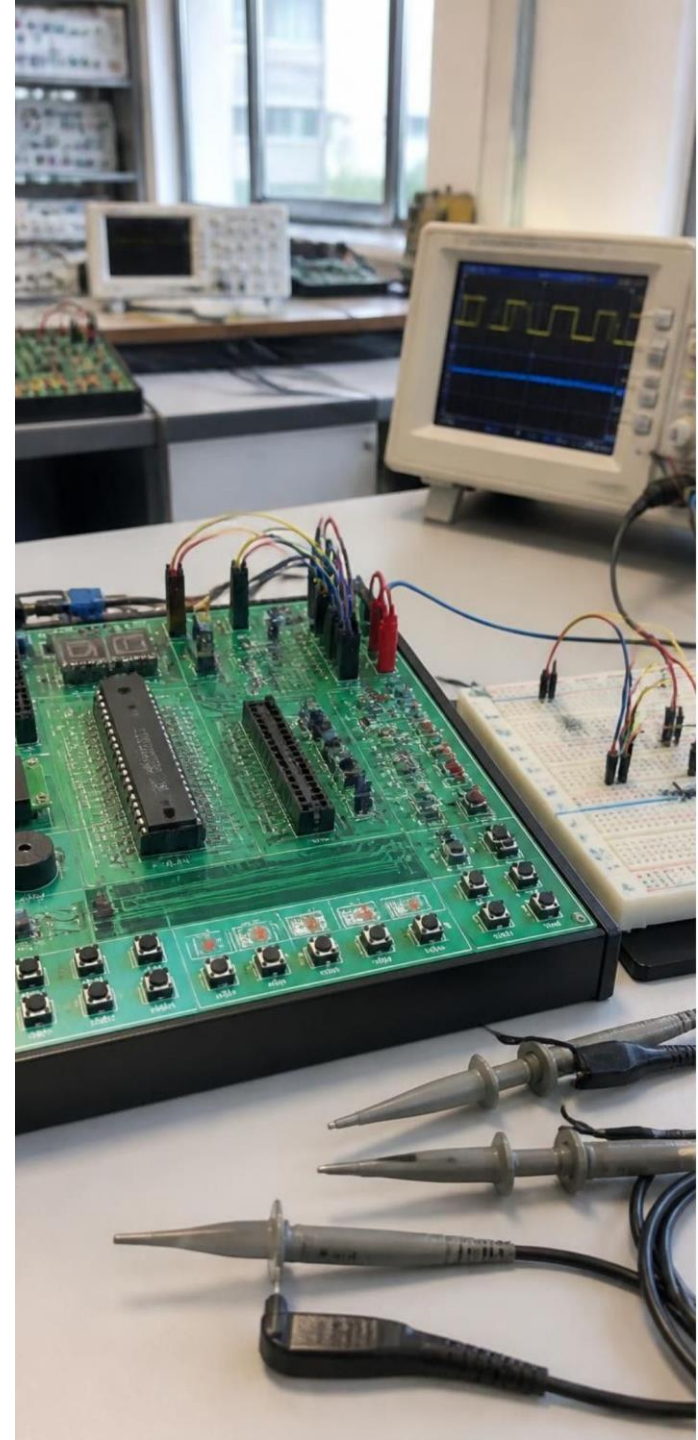
Advanced Computer Architecture, Contemporary Logic Design, Digital Electronics, VHDL and FPGA Programming, Microcontroller Programming.

AI and software

Artificial Intelligence with neural network implementation, C, C++, Python, OOP, Mobile Programming, Algorithm Design.

Systems

Database Engineering, Operating Systems, Software Engineering, Computer Applications in Management.



Education in Computer Systems Architecture

Ph.D. candidate

Computer Engineering (Computer Systems Architecture)

Azad University, North Tehran Branch

Master's

Computer Engineering (Computer Systems Architecture), same field and branch.

18.15 / 20 average · Thesis rated Very Good

Bachelor's

Computer Engineering (Software Engineering)

17.78 / 20 average

CONTACT

Let's collaborate



Academic inquiries

Academic collaborations, research inquiries and speaking.

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Via the university

Industry inquiries

Industrial automation, embedded design and software consulting.

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RESEARCH · ENGINEERING · TEACHING

Yahya Iranmanesh

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Hardware Processing for IQWG | FPGA | Embedded AI

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